

WASP-43b

R-2682

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_130406 · created 2026-08-30T23:53:42+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456388.71167 +/- 0.0005 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.016
Transit depth (Rp/Rs)^2	2.63 +/- 0.53 [%]
Orbital Inclination (inc)	82.14 +/- 1.32
Airmass coefficient 1 (a1)	0.951 +/- 0.012
Airmass coefficient 2 (a2)	0.035 +/- 0.037
Scatter in the residuals of the lightcurve fit is	1.26 %
Best Comparison Star	#3 - [34, 343]
Optimal Aperture	3.93
Optimal Annulus	12.24
Transit Duration (day)	0.0509 +/- 0.0045

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.016 vs 0.1594 (deviation 0.16σ)
Duration fit vs published	0.0509 d vs 0.0512 d (deviation 0.08σ)
Mid-time O–C	1.2 min
Depth SNR	10.1
Residual scatter	1.26 %

Light curve

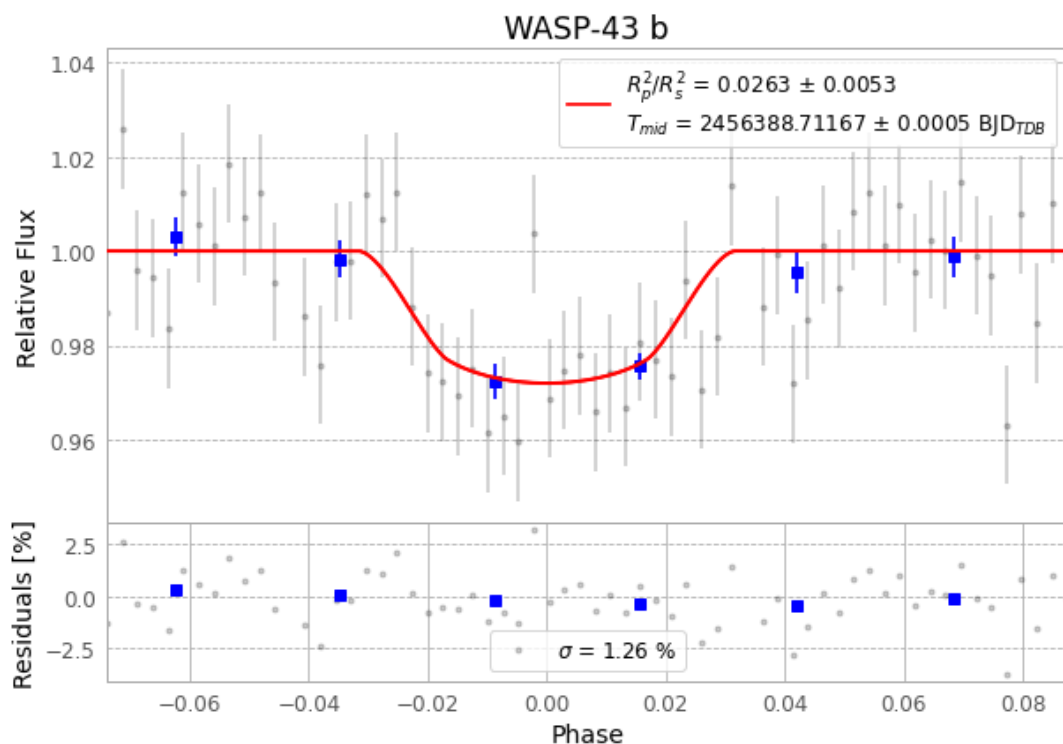


Figure 1 — FinalLightCurve_WASP-43 b_2013-04-05.png (SHA-256 270eb58f8ec51669...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [164, 299], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 939a7f600c0130be...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

64 FITS frames (42 MB), WASP-43130406033013.FITS → WASP-43130406063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-130406.log (32cddae5f784...), FinalLightCurve_WASP-43 b_2013-04-05.png (270eb58f8ec5...), FinalLightCurve_WASP-43 b_2013-04-05.csv (20471f7b7571...)

Compute resources

Wall time 287.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-30 23:53Z.